Waitlist COLA:
A Record-Blind NBA Draft Order With No Tanking Reward

Timothy Highley* Kaushik Rajan†

July 25, 2026

Abstract

A basketball team eliminated from playoff contention has a reason to keep losing. The worse it finishes, the better its position in the next draft of incoming players. In the 2025-26 National Basketball Association (NBA) season, by some counts a third of the league's thirty teams were openly doing this. The league assigns its top picks through a *draft lottery*, a weighted random draw among the fourteen teams that missed the playoffs, and has answered the problem by repeatedly adjusting the odds in that draw. Banchio and Munro [1] prove the approach cannot succeed. Any rule setting draft odds from end-of-season records, while giving a worse record better odds, must reward losing.

We show the odds are also the wrong place to look. Only the first four picks are drawn; picks 5 through 14 are handed out in strict reverse order of record, and most of the incentive lives there. Making the draw ignore records entirely, while leaving those ten alone, raises the reward for losing, from 1.30 to 1.98 places in the draft order. Removing records from the ten instead cuts it to 0.61.

Carry-Over Lottery Allocation (COLA) assigns draft priority from a multi-year index of how long a franchise has gone without success [2], with a single season's record playing no part. We formalize it as a configuration space of seven design dials, place each published variant and the league's current rule inside it, and evaluate nine mechanisms across 14,004 league-seasons in ZenGM Basketball-GM, an open-source simulator that plays every game with generated rosters, contracts, and automated general managers. Each is scored on one quantity read two ways, the gap in draft position between the team a ranking puts first and the team it puts fifth. Ranked by the season just played it is what losing bought; ranked by the mechanism's own definition of the most deserving team, it is how firmly the rule helps whom it means to help.

Wherever a rule calls the worst-record team the most deserving, those two rankings coincide and the two numbers are identical, which is the impossibility made visible. Today's lottery reads $+1.09 \pm 0.12$ in both. One configuration separates them, and we name it *Waitlist* COLA: the first four picks are drawn with odds proportional to the index, the remaining ten follow the index in descending order, and teams level on the index are separated at random. It holds the tanking reading at -0.12 ± 0.22 and the help reading at $+1.71 \pm 0.13$. Moving one team from the fifth-worst record to the worst, with the rest of the simulated season held fixed, buys 1.87 places under the original COLA variant and 1.32 under today's rule, and exactly zero under Waitlist COLA, in all 720 seasons and at every starting position.

The league has committed to revisiting its draft in 2029, and the change this implies is confined to picks no broadcast covers.

1 Introduction

Each summer the thirty franchises of the National Basketball Association (NBA) select from a pool of incoming players in a fixed order. Drafting early matters, because the difference between the first selection

*La Salle University. highley@lasalle.edu
†Independent researcher. kaushi@alumni.ncsu.edu

and the twentieth is often the difference between acquiring a franchise player and acquiring a substitute, and because a drafted player arrives on a fixed, below-market contract. The order in which teams pick is therefore one of the most consequential rules the league writes.

That order is set almost entirely by how badly a team performed. The sixteen teams that reach the playoffs pick last, in reverse order of finish. The fourteen that miss the playoffs pick first, and their order is decided by the *draft lottery*. The lottery is less of a lottery than its name suggests, and the detail matters for everything that follows. Only the first four picks are drawn at random, from a weighted pool in which a worse record earns more chances; the current weights give the three worst teams identical odds and decline from there. The remaining ten picks are not drawn at all. They are assigned in strict reverse order of regular-season record, so among teams that win nothing in the draw, the one that lost the most games picks ahead of the one that lost fewer.

This structure creates the incentive the league has spent a decade trying to remove. A team eliminated from playoff contention in February improves its expected draft position by losing its remaining games, a practice the sport calls *tanking*. It is not hypothetical or marginal. In the 2025-26 season, by some counts roughly a third of the league's thirty teams were openly losing at season's end to improve their position.

The league's responses have all worked on the same component. In 2019 it flattened the top of the odds table, equalizing the chances of the three worst teams so that finishing last no longer bought a distinctly better draw, and it expanded the drawn picks from three to four. In 2026 it took up a further proposal, referred to as *3-2-1*, which would widen the lottery to sixteen teams sorted into four tiers of four and allocate chances by tier in the ratio $2 : 3 : 2 : 1$ from the worst tier down [3]. The proposal deliberately gives the bottom tier fewer chances than the tier above it, and it admits the teams eliminated in the *play-in tournament*, the short elimination round in which the seventh through tenth finishers in each conference compete for the last two playoff berths. A three-year re-evaluation provision attaches to it, which reopens the design question in 2029.

There is a proof that this entire line of attack cannot succeed, and it was presented at this conference. Banchio and Munro [1] define a *No-Tanking Condition* (NTC), namely that no team can improve its expected outcome by losing, and show that any mechanism that maps end-of-season records to draft probabilities while giving worse records strictly better odds must violate it. The only escape within that family is a uniform lottery giving every non-playoff team identical odds, which satisfies the condition by abandoning the goal of helping the weakest teams first. Re-weighting trades one objective against the other; it does not resolve the conflict.

Our first empirical result is that the odds are also the wrong component to be arguing about. We decompose the reward for losing under today's rule into the part contributed by the weighted draw and the part contributed by the ten picks assigned by record, measured across simulated seasons. The draw is not where the reward lives. Making the draw ignore records completely, giving all fourteen teams equal chances at the first four picks while leaving the remaining ten in reverse-record order, does not reduce the reward for losing. It raises it, from 1.30 to 1.98 places in the draft order, because a record-weighted draw pulls the very worst teams out of the record-ordered queue and so compresses the gap that queue creates. Removing records from those ten picks instead, and leaving the draw exactly as it is today, cuts the reward to 0.61. A reform that flattens the odds while leaving the unlotteried picks alone is working on the smaller of two channels, and can make the problem worse.

Carry-Over Lottery Allocation (COLA) is a family of draft rules that escapes the impossibility by changing what the draft reads [2]. Each team carries an integer *lottery index* from season to season. A team that misses the playoffs adds to its index every year it keeps missing them, so the index measures how long a franchise has gone without success, on a horizon no single spring can move much. Success spends the index back down, since advancing in the playoffs cuts it by a factor that grows with how far the team went, and winning a top draft pick cuts it again, so no franchise can bank priority indefinitely or collect twice for one rebuild. Draft priority then follows the index. The manipulation channel behind the Banchio-Munro counterexample loses its grip, because one extra season of losing moves a multi-year quantity by a single year's accumulation, and moves it identically for every other team out of the playoffs.

COLA is a family, and its members differ on precisely the component our decomposition identifies. The original variant, which we call Classic COLA, draws the first four picks with odds proportional to the index and then orders picks 5 through 14 by reverse record, matching the shape of the lottery the NBA runs today. That tail is a concession to familiarity, and it is also where the entire remaining tanking reward sits. This paper studies the configuration that removes it. *Waitlist* COLA keeps the four-pick draw exactly as Classic runs it, then orders picks 5 through 14 by the index in descending order and separates teams level on the index at random. Teams that do not win the draw are served in the order they have been waiting, and nothing a team does on the court this season changes its place in that queue. A record still determines who enters the pool at all, which is the one manipulation channel that survives and the one we bound in §4.

Contributions:

- **(§5) The reward for losing is not in the odds.** Decomposing today's rule shows the weighted draw contributes less of the tanking incentive than the ten picks assigned by record, and that flattening the draw alone raises the incentive. Every reform of the last decade has adjusted the draw.
- **(§3) Seven-dial configuration space.** We formalize the COLA family as a 7-tuple $(E, \Delta, \rho, W, C, S, T)$, naming the seven independent choices any such rule must make: who is eligible, how much index a team gains per season, how much success takes away, how the index maps to draft positions, whether the index is capped, how long it persists, and how ties are broken (Definition 2). Each published variant is one point in this space, the 3-2-1 proposal sits at a corner where four of the seven dials are inert, and Waitlist COLA differs from the original variant on one dial. Because each dial maps to a property a league would care about, comparing rules becomes comparing tuples.
- **(§5) Full-engine evaluation of nine mechanisms.** We evaluate the family across 14,004 league-seasons in ZenGM Basketball-GM, an open-source simulator that plays out every game from generated rosters and contracts under automated general managers, alongside the NBA's current lottery and both readings of the 3-2-1 proposal. Scoring is one statistic read two ways, the gap in assigned pick between the team a ranking puts first and the team it puts fifth. The framing does real work. Wherever a rule's own notion of the most deserving team is the worst-record team, the two readings coincide by construction, so the rule cannot aim help without paying for losing. Waitlist COLA separates them, holding the tanking reading at zero while concentrating help as firmly as any configuration that also holds it at zero.
- **(§5) A causal price for losing, not an association.** A tanking gradient compares different teams in different situations, which measures a correlation. We add a paired counterfactual that holds one team fixed and moves only its record, so the quantity is the one a front office actually faces. Dropping the fifth-worst team to the worst record buys 1.87 picks under Classic COLA and 1.32 under today's NBA lottery. Under Waitlist COLA it buys exactly zero, in every season and at every starting rank, because the mechanism has nothing to read.
- **(§4, §5) Bounds on the channel that survives, analytic and measured.** A record still decides who enters the lottery pool, so a team can still throw games to a rival near the playoff boundary and change the field it competes against. We give a closed-form ceiling on the first-pick probability such a swap can buy (Theorem 2), a bound on what a team can gain by losing a playoff series under the capped variant of the family (Lemma 4), and Corollary 5, under which the value of that cap drops out of the bound and only the size of the eligible pool matters. We then measure the swap on the engine's

own seasons in the unit a general manager would use. Waitlist COLA concedes at most 1.06% of pick-one probability against the analytic ceiling, and at most 0.48 expected picks to the best-placed manipulator, which is what the current NBA lottery concedes on the same seasons.

2 Background and Related Work

2.1 Effort Manipulation: The Banchio-Munro Impossibility

Banchio and Munro [1] prove that any mechanism mapping end-of-season records to draft probabilities, while giving the worst team strictly better odds, must violate NTC. The proof constructs a minimal counterexample: two eliminated teams with identical records and a final game between them. The loser finishes with a worse record and therefore receives better draft odds; both teams prefer to lose. Only the uniform lottery satisfies NTC, but uniform odds abandon differential reward of the weakest teams.

The authors propose *T-IC* (Truncated Incentive Compatibility) as an alternative: track each team's conditional probability of finishing last and freeze allocations at the moment IC would first be violated. The mechanism is optimal in theory but has three practical issues. It is opaque to fans; the freeze point depends on team-level playoff-versus-draft valuations the league cannot observe; and the timing assumes teams begin the season with an incentive to win, when some teams tank from the first game.

2.2 Preference Manipulation and Axiomatic Limits

A second impossibility addresses preference misreporting (a team claiming to want a player it does not, to manipulate allocation order): Coreno and Balbuzanov [4] prove that no draft rule simultaneously satisfies Respect for Priority [5], Envy-Free up to One Object [6, 7], and Strategy-Proofness. The result is single-period. Whether the axioms lift to the multi-year priority structure of COLA is open and outside this paper's scope; we note the parallel impossibility as a complement to the Banchio-Munro effort-manipulation channel that motivates the bound in Section 4.

2.3 Prior Reform Proposals

Three proposals form the immediate context, and each accepts a different side of the tradeoff. The *uniform lottery*, in which all fourteen non-playoff teams hold identical odds, is the only rule in the record-reading family that satisfies the No-Tanking Condition, and it does so by abandoning any preference for the weakest teams. The league has not adopted it. The *wheel*, a pre-committed rotation in which each franchise's future draft positions are fixed years in advance regardless of performance (attributed informally to Boston Celtics executive Mike Zarren), removes tanking by severing record from draft order entirely, and with it removes the mechanism by which weak teams rebuild. The *3-2-1 lottery* described in §1 is structurally a uniform lottery with an adjustment for the play-in round and a reduction for the bottom three teams, which again buys the no-tanking property by giving up on helping the worst teams first. COLA takes a different route, shifting the basis of priority from a single season's record to multi-year playoff history, an axis none of the three considers.

3 The COLA Mechanism Family: A Configuration Space

COLA is a family of mechanisms parameterized by seven design dials. Each NBA lottery reform proposal under recent discussion (the status quo weighted lottery, the 2026 "3-2-1" proposal, and each COLA variant in Highley et al. [2] and the Substack series) corresponds to a specific choice of values for those seven dials. A policymaker who wants to move a downstream property (anti-tanking strength, weak-team rebuild horizon, broadcast complexity, multi-year predictability) is moving one or more dials. We develop the lottery index as the family's unified state variable (§3.1), the seven dials (§3.2), the variant map (§3.3), and the dial-to-property map (§3.4). Incentive compatibility rests on five assumptions [2], summarized in §3.5.

3.1 Lottery Index

Every COLA variant maintains a per-team integer state across seasons and uses it to determine draft priority (DP) in the current season. We call this state the *lottery index* and write $L_i^{(t)}$ for team i at the end of season t. Depending on the variant, $L_i^{(t)}$ tracks drought length (consecutive seasons without a playoff series win or top-3 pick) or a stockpile of lottery tickets; both are manifestations of the same index.

Definition 1 (Lottery Index). *For each team i and season t, the lottery index $L_i^{(t)} \in \mathbb{Z}_{\geq 0}$ is determined by an initial value $L_i^{(0)}$, a per-season* increment $\Delta_i^{(t)}$ *(variant-specific, depending on team performance and eligibility), and a per-season* diminishment $\rho_i^{(t)} \in [0,1]$ *that scales the index down based on playoff and draft outcomes:*

$$L_i^{(t+1)} = \left(L_i^{(t)} + \Delta_i^{(t)}\right) \cdot \left(1 - \rho_i^{(t)}\right). \tag{1}$$

The pool *in season t is $P^{(t)} = \sum_{i \in E^{(t)}} L_i^{(t)}$, where $E^{(t)} \subseteq \{1, \ldots, 30\}$ is the set of teams eligible for the lottery in that season.*

Three dials live in this update: the eligibility set $E^{(t)}$, the increment $\Delta_i^{(t)}$, and the diminishment $\rho_i^{(t)}$. A non-playoff team has positive increment (it accumulates priority); a team that wins a playoff series or a top pick has positive diminishment (it consumes some or all of that priority). §3.2 formalizes these three together with four additional dials that complete the parameterization, and §3.3 populates the 7-tuple for each variant.

3.2 Seven Dials

Four additional dials, implicit in the prose definitions of Highley et al. [2] and the Substack series but not yet formalized as design degrees of freedom, complete the configuration: how draft positions are assigned given the index vector, whether the index is bounded above, how long it persists across seasons, and how ties are resolved.

Definition 2 (COLA Configuration). *A* COLA configuration *is a 7-tuple $(E, \Delta, \rho, W, C, S, T)$ specifying:*

1. ***Eligibility** $E^{(t)} \subseteq \{1, \ldots, 30\}$: a season-$t$ predicate selecting the teams that participate in the lottery (per Definition 1).*
2. ***Increment** $\Delta_i^{(t)} \in \mathbb{Z}_{\geq 0}$: a non-negative integer added to the index of each eligible team, determined by season-t state (per Definition 1).*
3. ***Diminishment** $\rho_i^{(t)} \in [0,1]$: the fraction of the index removed after team i's playoff or draft outcome (per Definition 1).*
4. ***Lottery weighting** W: a rule mapping the index vector $L^{(t)}|_{E^{(t)}}$ to a distribution over assignments of teams to draft positions $\{1, 2, \ldots, |E^{(t)}|\}$. The rule may be deterministic (e.g., assign by index rank) or random (e.g., draw pick 1 with probability L_i/P, then draw pick 2 from the remainder).*
5. ***Cap** $C \in \mathbb{Z}_{>0} \cup \{\infty\}$: an upper bound on $L_i^{(t)}$, enforced as $L_i^{(t)} \leftarrow \min(L_i^{(t)}, C)$ after each increment. $C = \infty$ denotes no explicit cap.*

6. ***Carry-over scope*** *S: the temporal persistence rule, one of* single-season *(recompute $L_i^{(t)}$ from season-t state alone, no cross-season memory),* unbounded multi-year *(accumulate without limit),* bounded multi-year *(accumulate up to C), or* reset-on-event *(clear the index when a specified event occurs).*

7. ***Tiebreak*** *T: a total ordering on $E^{(t)}$ that resolves cases where $L_i^{(t)} = L_j^{(t)}$ for two or more eligible teams.*

A COLA variant *is a specification of all seven dials.*

Dials 4–7 are not new mechanisms. Each variant in Highley et al. [2] and the accompanying series already fixes a value for every one of them. Promoting them to typed dials enables cross-variant comparison via a single configuration object and dial-by-dial analysis of downstream properties. §3.3 populates the configuration per variant; §3.4 maps dials to league-relevant properties.

3.3 Variant Map

Six rules are worth locating in this space before the table states them formally, five from the COLA family and one from the league. *Simple* COLA does away with the draw entirely and orders the whole pool by drought. *Simple Lottery* COLA keeps that drought ordering but restores a weighted draw on top of it, using the odds the NBA itself used before 2019. *Classic* COLA is the original variant and the reference point for the rest of this paper, drawing four picks with odds proportional to the index and filling the remainder by reverse record. *Waitlist* COLA, introduced here, changes that remainder to index order. *Countdown* COLA scores teams by drought multiplied by wins, a quantity its authors call the McCarty number, and draws from nested pools so that no team can fall more than four places below its priority rank. *Capped* COLA bounds how much index any franchise can accumulate. The *3-2-1* proposal is included because the same seven dials describe it, and describing it that way shows what it gives up.

Table 1 states all seven dials for each. Cell values follow the specifications in Highley et al. [2], the accompanying essay series [8, 9, 10], and Highley [3].

The variants trade explainability, anti-tanking strength, and exposure to playoff-series manipulation against one another. Simple and Simple Lottery COLA appear in Highley [9]; Countdown COLA in the same series; and Capped COLA in Highley [10], introduced after the league was reported reluctant to adopt a lottery whose pool size could vary, and whose cap of $\text{MAX} = 150$ is what bounds the incentive to lose a playoff series (Lemma 4). The 3-2-1 proposal sits at a corner of the space where four dials are inert. Nothing accumulates, nothing diminishes, no cap binds, and priority is recomputed from scratch each season, leaving only eligibility and the draw itself to carry the design (§3.6). **[Q-HIGHLEY:** Beckett COLA is used as an empirical anchor in §5 but does not appear in Table 1; supply its 7-tuple to add it to the map, or we label it an off-family reference point alongside Countdown.]

Waitlist COLA is new here, and the formalism is what makes it easy to state. It holds every Classic setting and moves the second half of the W dial from reverse record to the index. That single move is what §5 shows carries the entire anti-tanking result, and it costs none of the structure that makes Classic legible, since the televised part of the event, the four-pick raffle, is untouched. The tiebreak dial then has to be filled in. Under Classic the tail is ordered by record and exact ties are rare and inconsequential; under Waitlist the tail is ordered by an integer index that increments in fixed steps, so several teams routinely arrive at the same value, and roughly a third of tail assignments in our runs involve at least one tie. Breaking those ties by record would smuggle the record back into the mechanism through the one door the design just closed, which is why T is a uniform random draw among equals. That choice is what makes the mechanism record-blind in the strict sense, ties included.

Dial	Simple	Simple Lottery	Classic	Waitlist	Countdown	Capped	3-2-1
E	22: non-playoff plus first-round losers	22 (same)	14: non-playoff only	14 (same)	22 (same)	rule-based exclusion	16 in four tiers
Δ	$+1$ per year	$+1$ per year	flat $\alpha = 1{,}000$	flat $\alpha = 1{,}000$	drought $\times$ wins	wins, for play-in teams and below	none
ρ	reset on a series win	reset on a series win	5 steps by playoff depth, plus draft	same as Classic	reset on advancing	6 steps by depth, plus draft	none
W	no draw, order by drought	pre-2019 NBA odds, then by drought	draw picks 1–4 at L_i/P, then **by record**	draw picks 1–4 at L_i/P, then **by index**	nested-pool draw, tickets $(6, 5, 4, 3, 2)$	draw picks 1–5 at L_i/P, then by record	draw by tier, balls $(2/3/2/1)$
C	none	none	none	none	none	$\text{MAX} = 150$ tickets	n/a
S	reset-on-event	reset-on-event	unbounded multi-year	unbounded multi-year	reset-on-event	bounded multi-year	single-season
T	most wins	no ties arise	no ties arise	**uniform at random**	no ties arise	most wins, among teams at the cap	no ties arise

Table 1: The seven dials, evaluated for each variant. Dial labels are E eligibility, Δ increment, ρ diminishment, W lottery weighting, C cap, S carry-over scope, T tiebreak (per Definition 2). Waitlist COLA differs from Classic COLA in two bolded cells, and the second follows from the first. Ordering the tail by the index makes exact ties common, so the tiebreak dial T, vacuous everywhere the draw is random, becomes load-bearing and has to be specified. The 3-2-1 proposal sits at a degenerate corner where Δ, ρ, and C are vacuous and S is single-season, leaving only E and W active.

3.4 Dial-to-Property Mapping

The configuration of Definition 2 reduces variant comparison to tuple comparison. The comparison is informative only when each dial corresponds to a downstream property. Four properties dominate the adoption discussion: anti-tanking strength (within-season manipulation foreclosed), weak-team rebuild help (how reliably the worst teams accumulate priority), complexity (legibility for fans and media), and predictability (multi-year forecastability for GMs and owners). Table 2 summarizes the mapping.

Three dials (E, Δ, C) bound the manipulator's pool share $p_{\max}$ and the size of any pool perturbation: a larger eligible set yields a larger total pool P, the increment Δ bounds per-team per-season accumulation, and the cap C bounds the index outright. Two dials (ρ, S) govern multi-year accumulation: diminishment determines how quickly an index decays after success; carry-over scope determines whether accumulation persists at all (the dial whose non-single-season value escapes the Banchio and Munro [1] impossibility). The remaining two (W, T) translate index dynamics into draft positions. A league that finds Classic too unbounded can move C to a finite value (the Capped move). A league that finds multi-year scopes administratively heavy can move S to single-season (the 3-2-1 move). The choice among variants becomes a choice among dial settings, not among monoliths.

Dial	Anti-tanking strength	Weak-team rebuild help	Complexity	Predictability
E	Sets the pool P in Theorem 2; narrower pools admit larger gain	Determines which weak teams accumulate priority	High if simple count, low if exclusion-rule	Stable if record-based; less so if exclusion-rule
Δ	Bounds the per-season marginal-decision gain	Flat α rewards uniformly; w-based rewards within-season effort	High if flat, lower if formulaic	High if record-based
ρ	Cliff size at each round transition; controls repeat-tanking returns	Restores priority to teams missing the playoffs again after success	Medium	Medium
W	Spread across rank positions; decouples position from index in part	Tilts allocation toward worst-record teams	High if deterministic, low if Monte Carlo	High if deterministic, lower if random
C	Bounds $p_{\max}$ and the per-series cost (Lemma 4)	Low cap compresses the priority signal	Medium	High once known
S	Escapes the Banchio and Munro [1] impossibility via multi-year scope	Multi-year scope rewards persistent weakness	Medium	Lower under multi-year scopes (legal/contracting)
T	Edge case for W-deterministic variants	n/a	High if simple	High

Table 2: Dial-to-property mapping. Cell entries describe the form the dial's influence takes; the dials and the variants of Table 1 together specify how to move from one league-relevant property to another by altering one or more dial values.

3.5 Incentive Compatibility

Highley et al. [2] establish incentive compatibility (IC) for Classic COLA under five assumptions:

1. **Playoff primacy:** a team's value of advancing in the playoffs exceeds any plausible draft-pick gain.
2. **Playoff advancement:** within the playoffs, advancing further is preferred to losing earlier, controlling for opponent strength.
3. **Picks 5–14 negligible at the margin:** no team profits from manipulating its position among those slots through within-season effort.
4. **Pool manipulation negligible:** no team can meaningfully shift its expected draft position by altering which other teams are in or out of the pool.
5. **No traded-pick protections:** pick trades do not interact with diminishment to create manipulation incentives.

Assumptions (1), (2), and (5) hold by construction or by short argument. Assumption (4) is supported in Highley et al. [2] by a 1,000-season simulation observing a maximum gain of roughly 3%, and §4 converts that observation into a closed-form bound.

Assumption (3) is the one that does real work, and it is the one Waitlist COLA discharges. Under Classic, picks 5 through 14 go by reverse record, so a team that loses one more game does move up one slot, and the assumption asks the reader to accept that the resulting gain is too small to act on. §5 measures that gain, and finds it substantial. Dropping from fifth-worst to worst is worth 1.87 picks. Under Waitlist COLA the same ten slots are ordered by a quantity no within-season choice can move, so assumption (3) becomes unnecessary. What remains is assumption (4). Waitlist inherits Classic's four-pick raffle unchanged, so the pool-swap bound of §4 transfers verbatim to the probability channel. It does not inherit the tail, and a deterministic index-ordered tail responds to pool composition in a way a record-ordered tail does not, which opens a second sub-channel that Classic does not have. §4.4 characterizes it and §5.3 measures it.

3.6 The 3-2-1 Proposal as a Degenerate Configuration

The NBA's 2026 "3-2-1" proposal, slated for an owners' vote at the end of May 2026, expands the lottery to sixteen teams across four tiers with ball counts $(2, 3, 2, 1)$ per tier. In the configuration of Definition 2: Δ, ρ, C are vacuous; S is single-season; E is the tiered 16-team pool; W is the tiered ball allocation; T is subsumed by the random draw. In Highley [3]'s framing, the mechanism is "essentially the uniform lottery but with an adjustment for the play-in and a punishment for the bottom 3 teams." Because the worst three teams get fewer balls than the middle seven, the proposal does not satisfy the premise of the Banchio-Munro counterexample (worse teams have strictly better odds), so it sidesteps the impossibility by abandoning the rationale for differential odds. COLA takes the opposite route, shifting the priority basis to multi-year playoff history through dial S.

4 Pool Manipulation Bound

A record decides who enters the lottery pool under every mechanism considered here, which leaves one channel open even when the draft order itself is record-blind. A team can lose deliberately to a rival sitting at the playoff boundary, push that rival into the playoffs, and change who it is competing against for picks. This section bounds what that is worth, treating the raffle and the deterministic tail separately, because they respond to a pool swap for different reasons and admit different kinds of bound. The raffle argument below applies unchanged to Classic COLA and Waitlist COLA, which share their first four picks exactly, and to any other configuration whose odds are a team's share of the pool total. §4.4 then treats the tail.

4.1 Setup

Let $P = \sum_{i=1}^{n} L_i$ denote the total lottery pool, where L_i is team i's lottery index and n is the number of lottery-eligible teams. A team's probability of winning the first pick is $p_i = L_i/P$.

The *pool manipulation* channel operates as follows. Team i deliberately loses to a high-index opponent h near the playoff boundary, causing h to make the playoffs. Team h (with index L_h) exits the lottery pool and is replaced by a lower-index team ℓ (with index $L_\ell < L_h$). The pool shrinks by $\Delta = L_h - L_\ell > 0$.

Proposition 1 (Manipulation Gain)**.** *The probability gain for team i from a pool swap of size $\Delta = L_h - L_\ell$ is $G_i = L_i \cdot \Delta/[P \cdot (P - \Delta)]$.*

Proof. Before manipulation, $p_i = L_i/P$. After the swap, the new pool is $P' = P - \Delta$ and $p'_i = L_i/P'$. The gain is $G_i = p'_i - p_i = L_i\big(1/(P - \Delta) - 1/P\big) = L_i \cdot \Delta/[P(P - \Delta)]$. □

When $\Delta \ll P$ (typically $\Delta/P \approx 4\%$ in our backtest), the gain factors into the manipulator's pool share (p_i) and the fractional pool change (Δ/P). Both have structural ceilings, which we exploit below.

Indexing convention. The bound uses $L_i = T_i \cdot \alpha$, where T_i is team i's consecutive non-playoff seasons. For Classic COLA and Waitlist COLA, which share the increment and diminishment dials and so share the index dynamics entirely, this is exact for any team starting from a stockpile reset (championship or top-5 lottery pick) and is the steady-state attractor under the diminishment schedule. A general bound handling ticket inheritance under the full diminishment dynamics is future work.

4.2 Conditioned Upper Bound

Definition 3 (Structural Anti-Correlation)**.** *Let $T_{\max}$ denote the maximum consecutive non-playoff years for any team, and $T_{boundary}$ the maximum consecutive non-playoff years for a team near the playoff boundary (a team that could plausibly be pushed into the playoffs by a single loss). The index distribution exhibits* structural anti-correlation *when $T_{boundary} \ll T_{\max}$: teams with the highest indices are far from the playoff boundary and thus cannot be the target of pool manipulation.*

Theorem 2 (Conditioned Manipulation Bound)**.** *Under structural anti-correlation with parameters $T_{\max}$, $T_{boundary}$, and average index multiplier $\bar{k} = \bar{L}/\alpha$, where α is the annual index increment, the maximum manipulation gain satisfies*

$$G_{\max} \leq \frac{T_{\max} \cdot (T_{boundary} - 1)}{(T_{\max} + n - 1) \cdot n \cdot \bar{k}}.$$

Proof sketch. The gain is maximized when L_i is maximized and Δ/P is maximized. *Bounding $p_{\max}$*: the highest-index team has $L_{\max} = T_{\max} \cdot \alpha$. The minimum pool size occurs when one team has $T_{\max}$ years of accumulation and the remaining $n-1$ teams each have the minimum α: $P_{\min} = (T_{\max} + n - 1) \cdot \alpha$. Thus $p_{\max} \leq T_{\max}/(T_{\max} + n - 1)$. *Bounding Δ/P*: the team being pushed out has at most $T_{\text{boundary}} \cdot \alpha$ tickets, and the team replacing it has at least α, so $\Delta_{\max} = (T_{\text{boundary}} - 1) \cdot \alpha$. In steady state, $P \approx n \cdot \bar{k} \cdot \alpha$. Combining via the first-order approximation yields the stated bound. □

Empirical calibration: We instantiate with $T_{\max} = 10$ (typical-case ceiling on consecutive non-playoff years across teams in our 26-season backtest), $T_{\text{boundary}} = 5$ (conservative ceiling for a near-boundary team), $n = 14$, and $\bar{k} = 3.1$ (the original paper's observed carry-over). These yield

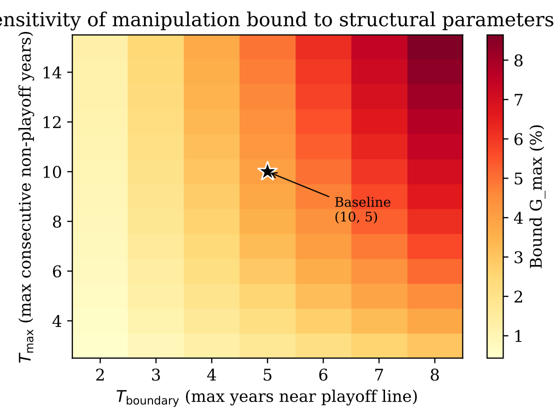
$$G_{\max} \leq \frac{10 \cdot 4}{23 \cdot 14 \cdot 3.1} \approx 4.0\%,$$

which contains the simulation-observed maximum of 3.0% across 1,000 seasons [2]. An independent 200-season simulation we ran, in which game outcomes follow a Bradley-Terry model (each matchup resolved by the two teams' relative strengths), confirms the bound at a maximum gain of 1.70%. The 26-season dataset contains one outlier (the Sacramento Kings' 16-year drought, 2006–07 through 2021–22). Setting $T_{\max} = 16$ loosens the bound to roughly 5.1%, still above the empirical extremes; we report the typical-case bound ($T_{\max} = 10$) as the headline because the next-longest droughts are Minnesota at 13 years and Phoenix/Charlotte at 10.

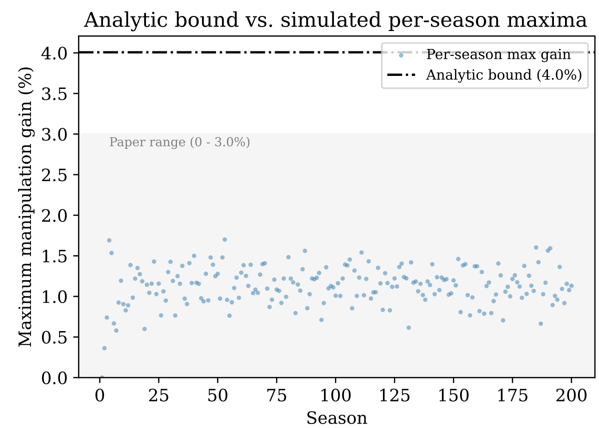
4.3 Sensitivity, Coalitions, and the Pick-Value Gap

Figure 1a shows T_{boundary} as the dominant lever. Under pessimistic assumptions ($T_{\text{boundary}} = 7$), the bound stays under 7%; under realistic conditions ($T_{\text{boundary}} \leq 5$), under 5%. T_{boundary} is also the empirical quantity most amenable to direct measurement.

Coalitional manipulation does not amplify individual gains. Joint gain for a coalition $\{A, B\}$ is $(L_A + L_B) \cdot \Delta/[P(P - \Delta)]$. At 30% joint pool share, joint gain remains $\leq 1.3\%$ and per-member gain cannot



(a) Bound vs. $T_{\max}, T_{\text{boundary}}$.



(b) Analytic bound vs. simulation.

Figure 1: Manipulation bound. (a) Sensitivity to the structural-anti-correlation parameters; the bound is most sensitive to T_{boundary}, and stays under 5% for $T_{\text{boundary}} \leq 5$. (b) Analytic bound (red) vs. the empirical distribution of manipulation gains across 200 Bradley-Terry-simulated seasons; the bound contains all simulated outcomes with margin.

exceed the highest-index individual's. Coordination costs (detection, trust, sequential uncertainty) further dampen the channel.

The bound addresses the *probability* term of the manipulation incentive. The *value* term (the pick-value gap $d_k - d_{k+1}$) has no formal ceiling in the original specification, though it is bounded for any realistic draft class. Fully closing assumption (4) would also require a bound on $d_k - d_{k+1}$ via the distribution of draft-eligible prospect quality; we leave this to future work.

4.4 The Tail Channel Under a Deterministic Ordering

Theorem 2 bounds what a pool swap buys in the raffle, where a team's stake is its share of the pool total. Waitlist COLA inherits that raffle unchanged, so the bound transfers with no modification. It does not inherit the tail, and a deterministic ordering responds to pool composition in a way a random draw does not. Remove a long-drought team from the pool and everyone below it in the queue moves up a place. Nothing in that argument is peculiar to the index. A tail ordered by reverse record answers pool composition the same way, and the NBA's current lottery orders its tail by reverse record, so the channel is neither introduced by Waitlist COLA nor avoided by keeping the status quo. The following bounds it for any deterministic tail at once.

Proposition 3 (Tail Displacement)**.** *Let the picks beyond the raffle depth be assigned by a deterministic total order on a key k over the lottery pool, with higher k picking earlier. Suppose the pool changes by the removal of one team and the addition of one other, and condition on the set of raffle winners among the teams present in both pools. Then every remaining team's tail position changes by at most one pick, in either direction.*

Proof. Conditional on the raffle winners, the tail position of team i is $1 + |\{j \neq i \text{ in the tail} : k_j \succ k_i\}|$. Removing one team decreases that cardinality by one if the departing team outranked i and leaves it unchanged otherwise; adding one team increases it by one if the arriving team outranks i and leaves it unchanged otherwise. The two adjustments are each at most one and carry opposite signs, so their sum lies in $\{-1, 0, +1\}$. $\square$

Two consequences matter. The exposure is bounded by construction, where Theorem 2's ceiling depends on the shape of the index distribution. And the same bound already applies to the mechanism the league runs today, so comparing Waitlist COLA to the status quo on this channel is a question of which teams fall inside the displaced band, not of whether the band exists. Under a record-ordered tail the displaced teams are those whose record falls between the team pushed into the playoffs and the team pushed out; under an index-ordered tail they are the teams whose index falls between the two. A manipulator cannot arrange to be among them, since the band is fixed by the records and indices of two teams it does not control. Set against this, the channel Waitlist COLA closes is worth up to ten picks, is open to any team in the pool, and requires nothing but losing games it is already playing. §5.3 measures both.

4.5 Per-Series Bound for Capped COLA

Capped COLA adds a fourth manipulation channel, namely tanking through the playoffs, where a team already at the cap declines to advance in order to keep its stockpile. The cap converts the playoff-diminishment schedule from a fraction-of-current-stockpile cost (unbounded as stockpile grows) to a fraction-of-MAX cost (bounded by construction).

Lemma 4 (Per-Series Tanking Cost)**.** *Let* MAX *be the Capped* COLA *stockpile cap. For any team entering the playoffs with* $L_i \leq \mathrm{MAX}$*, the maximum reduction in* L_i *attributable to winning (rather than losing) any single playoff series is bounded by:*

$$\Delta L^{\max}_{\textit{series}} = \begin{cases} 0.3 \cdot \mathrm{MAX} & \textit{play-in advancers in round 1,} \\ 0.2 \cdot \mathrm{MAX} & \textit{any other series advancement.} \end{cases}$$

At $\mathrm{MAX} = 150$*, the bound is* 45 *tickets for the play-in case and* 30 *tickets for any subsequent series.*

Proof. Reading off Capped COLA's six-step diminishment schedule, the per-series increments in the diminishment fraction $\Delta\rho$ are: play-in advancer round 1 ($\rho_{\text{lose}} = 0.1$, $\rho_{\text{advance}} = 0.4$, $\Delta\rho = 0.3$); standard round 1 ($0.2 \to 0.4$, $\Delta\rho = 0.2$); round 2 to conference finals ($0.4 \to 0.6$); conference finals to NBA finals ($0.6 \to 0.8$); finals to championship ($0.8 \to 1.0$). All non-play-in transitions yield $\Delta\rho = 0.2$. The cost in tickets is $\Delta\rho \cdot L_i^{\text{pre}} \leq \Delta\rho \cdot \mathrm{MAX}$, with equality at the cap. □

The bound makes the playoff-tanking incentive transparent. At $\mathrm{MAX} = 150$, no single series win ever costs more than 45 tickets, roughly a year and a half of accumulation for a team near the wins-increment ceiling. The cumulative regret for a finals-losing team at the cap is $0.8 \cdot \mathrm{MAX} = 120$ tickets, decomposed per Lemma 4.

Corollary 5 (Cap-Cancels-Out for Capped Configurations)**.** *For a Capped* COLA *configuration with eligibility pool* E *of size* $|E|$*, cap value* C*, and per-series-cost upper bound* $\kappa \cdot C$ *(where* $\kappa = 0.3$ *in the play-in worst case of Lemma 4 and* $\kappa = 0.2$ *for any subsequent series), the worst-case manipulation-gain bound for a single-series pool perturbation is*

$$\Delta p_{\textit{capped}} \;\leq\; \frac{\kappa \cdot C}{C \cdot |E|} \;=\; \frac{\kappa}{|E|}.$$

The cap value C *drops out. The manipulation-gain bound depends only on* κ *and* $|E|$*.*

Proof. The maximum index a single team can hold under Capped COLA is C, and the worst-case pool of $|E|$ such teams is $C \cdot |E|$. The maximum single-series perturbation in a team's index from the pool is $\kappa \cdot C$ tickets, by Lemma 4. Substituting into the gain expression $\Delta = (\text{perturbation})/(\text{pool})$ yields $\kappa \cdot C/(C \cdot |E|) = \kappa/|E|$. □

The cap controls per-team accumulation speed and per-series tanking cost (Lemma 4); it does not control pool-wide manipulation resistance, which is set by $|E|$. A policymaker can tune C on accumulation-speed and per-series-cost preferences without affecting the manipulation-gain ceiling. At Capped's published defaults ($|E| = 22$, $\kappa = 0.3$ play-in or 0.2 standard), the bound is $\Delta p_{\text{capped}} \leq 0.3/22 \approx 1.36\%$ play-in and $0.2/22 \approx 0.91\%$ for any subsequent series.

4.6 Three Closures of Assumption (4)

The pool-manipulation gap admits three closure strategies. *Empirical:* Highley et al. [2] simulate 1,000 Bradley-Terry seasons and observe maximum gain $\sim 3\%$ (statistical, no parameter-explicit ceiling). *Behavioral:* the companion formal-proof video to [2] introduces $G >$ expected pool-manipulation gain (where G is the value of a regular-season win), eliminating the channel via preference structure. *Analytic (this paper):* Theorem 2 gives a closed-form bound conditioned on structural properties of the index distribution, parameter-explicit and admitting sensitivity analysis (Figure 1a). The three compose; any practical deployment can rely on whichever best matches the available evidence.

5 Empirical Validation

We instantiate the configuration of §3 in the full ZenGM Basketball-GM engine. §5.1 discloses the simulation model and why it is the right reference for a draft-reform question. §5.2 describes the nine mechanisms, the three legs they are scored on, and the counterfactual that turns the tanking leg from an association into a causal price. §5.3 reports the results together with the run's limits, and §5.4 carries the 26-season historical backtest.

5.1 Simulation Model

The ZenGM Basketball-GM engine implements all five COLA variants and the standard NBA lottery; we use it as our reference implementation and drive it from a script with no graphical interface, leaving the full per-game simulation active. The engine generates rosters and contracts, simulates each game, lets AI general managers draft and trade, and applies city-size effects; team strength emerges from the rosters the engine actually plays, not from a hand-specified law of motion. This is the realism the draft-reform question requires. Tanking, rebuild help, and pool manipulation are all consequences of how a drafted player changes the team that drafts him, so the channel from draft pick to standings has to be the engine's own, not a stipulated one. The engine's own lottery code is left unmodified, so draws and per-team index updates run exactly as the public implementation performs them, and the mechanisms under test are introduced by configuring that code.

Three things support reading these results as evidence about basketball and not only about a simulator. The draft-to-outcome channel is verified inside the engine before it is relied upon (§5.3), so the link from an earlier pick to later wins is measured. The pick distributions used for the counterfactual and manipulation analyses are checked against the engine's own draws before use. And the historical backtest of §5.4 applies the same rules to 26 seasons of real NBA results, where the qualitative comparative statics reappear.

We report uncertainty at the replicate level (effective sample size 48 per mechanism; see the Simulation-parameters note), so within-league season correlation does not inflate significance. Each claim carries a 95% confidence interval, where a near-zero claim rests on an interval that includes zero and a discriminating claim on one that excludes it.

The counterfactual and manipulation legs need an exact model of each mechanism's pick distribution, since both are differences between two scenarios that often differ by a fraction of a pick, and sampling both sides would put noise on a quantity smaller than the noise. Every mechanism here has the same two-stage shape, a raffle without replacement followed by a deterministic ordering, so the distribution enumerates in closed form. We validated it against the engine on three tests before use. The pool it scores is exactly the set of teams the engine handed picks 1 through 14, in all 720 seasons of every arm. Conditional on the raffle winners the engine actually drew, the remaining picks are deterministic, and the model reproduces all 21,600 of them, counting a team tied on the ordering key as correct anywhere inside its tie block. And its exact pick-one probabilities match the frequencies the engine realized within sampling error in every bin of every arm, the largest deviation being 1.5 standard errors.

Scale and composition of the runs. Each mechanism is evaluated over 48 independent leagues, seeded per replicate and simulated for 15 consecutive seasons, and uncertainty is reported at the replicate level so correlation between seasons of the same league does not inflate it. The effective sample size for any mechanism comparison is therefore 48, well below the raw season count. The study spans 14,004 league-seasons (420,120 team-seasons) across four runs. A channel probe (12 leagues) establishes what a draft pick is worth. An initial sweep over seven mechanisms at a 22-team eligibility pool (7×48 leagues) located the lottery-weighting dial as the one that matters. The headline sweep re-runs six configurations at the 14-team pool that Classic COLA specifies (6×48 leagues), and a reference sweep adds the NBA's current lottery and both readings of the 3-2-1 proposal (3×48 leagues). A paired tanking experiment contributes a further 6×48 leagues of 8 seasons. The pool-manipulation and counterfactual-tanking analyses re-use these runs and add no seasons of their own.

5.2 Testbed Setup

A channel probe (§5.3) first establishes what a draft pick is worth in the engine, so the rest can be read against the right magnitude. The main experiment holds every dial of Definition 2 at its Classic setting except the lottery weighting W, and fixes eligibility at Highley's canonical 14-team pool for every arm. Two things vary. The first is the weighting exponent, since the draft order over the pool is drawn with probability proportional to the index raised to a power L_i^γ, where $\gamma = 0$ gives every pool team equal odds, $\gamma = 1$ is proportional to drought, and larger γ concentrates odds on the longest-suffering. The second is the raffle depth, meaning whether the proportional draw assigns every pick or assigns the top four and hands the remainder to a deterministic ordering. Classic COLA and Waitlist COLA both stop the raffle at four and differ only in what orders the remainder, which makes the pair a controlled comparison of the tail rule on its own.

Three record-based mechanisms complete the field, so the COLA configurations are scored against what a league would actually be choosing between. The NBA's current lottery is implemented with its official post-2019 ball counts (140/140/140/125/105/90/75/60/45/30/20/15/10/5 per thousand by record rank), a top-four draw, and picks 5 through 14 by reverse record. The 2026 3-2-1 proposal is implemented twice, because its public description does not settle how deep the tiered draw runs, with one arm using an NBA-style top-four draw over the tiered balls and the other drawing the full pool. Nine mechanisms run in total, each over 48 independent leagues of 15 consecutive seasons. The multi-year index accumulates and decays under the same law in every arm, including the three that never consult it, so the drought covariate means the same thing everywhere. One implementation detail is worth stating, since Waitlist COLA specifies a random tiebreak and the engine does not implement one. Teams tied on the index are separated in the runs by the team identifier assigned at league creation. Within the lottery pool that identifier is very nearly independent of record (correlation 0.003 across 8,064 team-seasons), but it is not perfectly so: the worst-record pool team carries a lower mean identifier than the fifth-worst (13.59 against 14.65), because the engine's team ordering tracks market size and market size tracks strength. The residual matters only where identifiers decide an outcome, which under Waitlist COLA is confined to exact ties on the index, and it cannot affect the counterfactual of §5.3 at all, since that intervention holds every identifier fixed and moves only a record. We quantify what an identifier-ordered rule would contribute on its own below.

Scoring follows the testbed specification, which reduces the first two legs to a single statistic read under two rankings. Rank the fourteen pool teams, take the team ranked first and the team ranked fifth, and report the gap in the draft pick they received. *Anti-tanking (leg a)* ranks by the season just played, so the gap is what a worse record bought, and the target is zero. *Help (leg b)* ranks the same teams by the mechanism's own priority metric, which is the record for the three record-based arms and the multi-year index for the COLA arms, so the gap is how hard the mechanism concentrates early picks on the teams it judges most deserving, and wider is better. Both are reported over seasons 4 through 15. The first three seasons of a run hold every index near zero, so no drought-based mechanism can separate teams that have not yet accumulated a drought; leg (a) is insensitive to that choice and leg (b) is not. A regression of assigned pick on accumulated drought-years accompanies leg (b) as a higher-powered estimator of the same question.

Both legs score where a team *picks*. Whether an early pick actually returns a franchise to contention is a further question, and we score it separately with two outcome measures. We treat reaching the conference finals, the round of four that decides the two teams playing for the championship, as the bar for contention, since it is reached by a small enough set of teams each year to discriminate and is not so rare that fifteen seasons yield too few events. The *drought tail* is the 90th percentile of years since a team's last conference-final appearance, where lower is better, and the *recovery rate* is the share of team-seasons already three or more years past a conference final that reach one within the next three seasons, where higher is better.

Leg (a) as defined compares different teams, so we add a paired counterfactual on the same runs. Holding a season exactly as the engine produced it, one team's record is moved and nothing else is, and the exact pick distribution is recomputed on both sides. Because the carry-over index updates on playoff outcomes alone (§3.1), a lottery team that loses more games carries an identical index in both arms, so a record is the only thing the intervention moves, which is also the only thing a within-season tank can move.

Manipulation (leg c) is measured. On every season, we apply the boundary pool swap of §4, pushing the best-record non-playoff team into the playoffs and the worst-record playoff team out into the pool, and recompute exactly. We report the gain in pick-one probability, which Theorem 2 bounds, and the gain in expected pick, which a front office would weigh and in which the tail channel of §4.4 is visible at all. A manipulator is credited with the best conference and the best-placed beneficiary in that season, so the figure is an envelope over a season's opportunities. Two conservatisms run against the record-based mechanisms. The manipulator's own record is held fixed, though the games it throws would also drop it down the standings, and the gain is credited even where it is worth less than the games given up.

Per-leg quantities are aggregated as the mean across the 48 leagues with the standard error of that mean, so correlation between seasons of the same league does not inflate significance.

5.3 Results

The draft channel is modest. The channel probe assigns picks at random and measures the effect on later standings. A first-overall pick is worth about two to three wins over a thirtieth pick at its peak ($n = 12$ leagues; the all-teams estimate sits about 5.2 standard errors from zero, the weak-team estimate about 4.6), and the assigned pick stays orthogonal to current record, so the effect is causal but small. This sets the scale for everything below. Which team a mechanism helps moves at the pick level; how far that help carries to the standings is limited by what one pick is worth.

The reward for losing is not where the reforms have looked. Today's lottery has two components, a record-weighted draw for the first four picks and a record-ordered queue for the remaining ten, and every reform of the last decade has adjusted the first. Table 3 switches each component off in turn, on the seasons the engine produced, and reports what the worst-record team gains over the fifth-worst in each case.

what the rule reads	worst vs fifth-worst	share of today's reward
today's rule, both components read record	1.30	100%
draw made record-blind, queue by record	1.98	152%
queue made record-blind, draw as today	0.61 ± 0.09	47%
neither component reads record	-0.16 ± 0.12	noise floor

Table 3: Decomposition of the current NBA lottery's tanking reward, 48 independent leagues, seasons 4–15, computed as exact expectations on the engine's realized seasons. The quantity is the expected-pick gap between the worst-record and fifth-worst-record team. The first two rows carry no standard error because they depend only on the rule's rank structure, which is identical in every season. A record-blind ordering is generated by a deterministic hash of season and team, since team identifiers themselves carry a faint record signal at these two order statistics.

Switching off the draw does not help. Giving all fourteen teams equal chances at the first four picks, while leaving the ten-pick queue ordered by record, moves the reward from 1.30 to 1.98 places, so a flat draw is worse for tanking than the weighted one it would replace. The reason is visible in who is standing in the queue. A record-weighted draw removes the very worst teams from the queue more often than it removes anyone else, which promotes every team behind them and compresses the gap the queue creates; flattening the draw stops doing that. Switching off the queue instead, and leaving the draw exactly as it is, cuts the reward to 0.61. Switching off both leaves -0.16 ± 0.12, consistent with the zero it must equal by construction and a useful floor: differences of a tenth of a pick are noise here, and the ones above are not.

The consequence for policy is direct. A reform that reweights the draw and leaves the unlotteried picks alone is working the smaller of two channels and may move the larger one the wrong way.

Two tests, one statistic (legs a and b). Table 4 scores all nine mechanisms. Both columns report the same quantity, the gap in draft pick between the team a ranking puts first and the team it puts fifth. Column (a) ranks by the season just played, so it measures what losing bought. Column (b) ranks by the mechanism's own priority metric, so it measures how hard the mechanism concentrates early picks on the teams it judges most deserving.

The first thing to read is that for every record-based mechanism the two columns are the same number. That is an identity, not a pattern in the data. When a rule's notion of the most deserving team is the team with the worst record, the ranking that scores help and the ranking that scores tanking are one ranking. Such a mechanism cannot aim help without paying for losing, whatever its odds table looks like, which is the Banchio-Munro impossibility appearing as a column of repeated entries. The NBA's current lottery posts $+1.09$ in both. The 3-2-1 proposal posts $+1.76$ under a top-four draw and -1.22 under a full-pool draw, and the sign flip is the whole design. Giving the bottom tier fewer balls than the tier above it buys a losing penalty by giving up on helping the worst teams first, in both columns at once.

Only a mechanism whose priority basis is something other than a record can split the columns, and the COLA configurations do. Waitlist COLA holds column (a) at -0.12 ± 0.22, an interval containing zero, while posting $+1.71 \pm 0.13$ in column (b). Classic COLA splits them too, but in the wrong proportion, leaving $+1.88 \pm 0.11$ of tanking reward for $+1.38 \pm 0.24$ of help. That Classic tanks worse than the lottery it would replace is the same effect Table 3 isolates. Both order picks 5 through 14 by reverse record, so both pay for losing in the same place, and only the NBA rule also weights its draw by record, which pulls the worst teams out of the queue and compresses the gap. Under Classic the draw is weighted by drought, which thins the queue without regard to who is at the bottom of it, so the queue's ordering is paid in full. A record-weighted draw masks a record-ordered queue, and moving the draw to a fairer basis while leaving the queue alone exposes what the queue was doing all along.

mechanism	(a) ranked by record want 0	(b) ranked by own metric want wide	6-year vs 1-year drought, picks
NBA lottery, 2019 rules	$+1.09 \pm 0.12$	$+1.09 \pm 0.12$	0.3
3-2-1, top-four draw	$+1.76 \pm 0.12$	$+1.76 \pm 0.12$	0.5
3-2-1, full-pool draw	-1.22 ± 0.25	-1.22 ± 0.25	0.2
Classic COLA	$+1.88 \pm 0.11$	$+1.38 \pm 0.24$	2.9
COLA, flat weighting	$+0.04 \pm 0.23^{*}$	$+0.30 \pm 0.29^{*}$	0.0
COLA, full-depth proportional	$-0.14 \pm 0.26^{*}$	$+0.82 \pm 0.20$	4.0
COLA, steep weighting	$+0.11 \pm 0.22^{*}$	$+1.77 \pm 0.17$	6.0
COLA, steepest weighting	$-0.26 \pm 0.25^{*}$	$+2.43 \pm 0.16$	7.3
Waitlist COLA	$\mathbf{-0.12 \pm 0.22}^{*}$	$\mathbf{+1.71 \pm 0.13}$	**6.6**

Table 4: Nine mechanisms at $E = 14$, 48 independent leagues $\times$ 15 seasons each, scored over seasons 4–15 (means $\pm$ standard error across leagues; a star marks a 95% interval containing zero). Columns (a) and (b) are the same statistic under two rankings, and are identical by construction wherever a mechanism's own priority metric is the record. The last column is the plain-units version of (b), the mean pick separating a team six or more years from a conference final from one that is one or two years removed. Column (a) is insensitive to the season window; over all 15 seasons it reads $+1.94$ for Classic COLA, $+1.17$ for the NBA lottery, and -0.07 for Waitlist COLA.

How much help, and measured how. The last column of Table 4 puts column (b) in units a general manager would use. Under the lottery the NBA runs today, a team six or more years removed from a conference final drafts 7.4th on average and a team one or two years removed drafts 7.7th, so length of suffering is worth a third of a pick. Under Waitlist COLA the same comparison is 4.4 against 10.9.

Two estimators of leg (b) are worth separating, because they disagree on how confidently Waitlist COLA beats Classic COLA and the disagreement is instructive. The two-team gap in column (b) is the testbed's own statistic and the most interpretable one, but it reads only two of the fourteen pool teams per season, and at that resolution Waitlist's $+1.71$ against Classic's $+1.38$ is not a separation (Welch $t = 1.22$). Regressing assigned pick on accumulated drought-years uses all fourteen, and there Waitlist moves a team 1.26 ± 0.03 picks earlier per drought-year against Classic's 0.99 ± 0.04, which is a clear separation ($t = 5.70$, $d = 1.16$). The two agree in direction and differ in power. We report the gap as the headline because it is the statistic the design question is posed in, and the slope as the evidence that the difference is real.

What a losing streak actually buys (leg a, causally). Column (a) compares different teams, so it cannot by itself say what a given team gains by losing. Table 5 answers that directly. On each of the 720 seasons per mechanism, one team's record is moved to the worst in its league while every roster, every index, and every other team's record is held at what the engine produced, and the exact pick distribution is recomputed on both sides.

Moving the fifth-worst team to worst is worth 1.87 picks under Classic COLA and 1.32 under the current NBA lottery. Under Waitlist COLA it is worth 0.000. That zero is not a small effect that failed to reach significance and not an artifact of averaging. Across every starting rank and all 720 seasons, in 18,720 separate interventions, the largest absolute change found anywhere is exactly zero, because the mechanism reads a quantity the intervention does not touch. The counterfactual also prices the reward at every size. A team already fourteenth in the pool that runs its record to the bottom gains 9.96 picks under the NBA rule and 6.69 under Classic, and the marginal value of one more rank of losing rises as a team falls, peaking near two full picks under the current lottery.

mechanism	5th-worst → worst	one more rank of losing	14th → worst
NBA lottery, 2019 rules	$+1.32$	$+0.70$	$+9.96$
Classic COLA	$+1.87$	$+0.70$	$+6.69$
Waitlist COLA	$\mathbf{0.00}$	$\mathbf{0.00}$	$\mathbf{0.00}$

Table 5: Picks bought by losing, paired counterfactual on 720 engine seasons per mechanism. One team's record moves and nothing else does, and both arms are computed in closed form, so the pairing is exact and the zeros are exact. The middle column is the marginal incentive at the fifth-worst position, which is what a team weighs when it decides whether to sit a healthy starter. Waitlist COLA's zero holds at every starting rank in all 18,720 interventions.

	expected picks gained			max gain in
mechanism	mean	max	seasons with a gain	$\mathbb{P}$(pick 1)
NBA lottery, 2019 rules	0.49	1.22	58%	1.50%
Classic COLA	0.34	1.08	86%	1.90%
Waitlist COLA	0.48	1.08	74%	**1.06%**

Table 6: Pool manipulation measured on 720 engine seasons per mechanism. A manipulator throws games so the best-record non-playoff team climbs into the playoffs, displacing the worst playoff team into the pool. Each season is credited with the best conference and the best-placed beneficiary available, so these are envelopes over a season's opportunities, and the manipulator's own record is held fixed, which understates the record-based mechanisms. The Theorem 2 closed form on the same seasons gives 3.98% for Classic COLA and 2.78% for Waitlist COLA; it does not apply to the NBA lottery, whose raffle is weighted by record rank.

The surviving manipulation channel, measured (leg c). A record still decides who enters the pool, so we apply the boundary swap of §4 to every engine season (Table 6). On the probability channel Theorem 2 bounds, Waitlist COLA concedes at most 1.06 percentage points of pick-one probability, against the closed form's 2.78% on the same seasons and the roughly 4% analytic ceiling, and it concedes less than Classic COLA does. The theorem's structural-anti-correlation premise holds in the engine, because teams with long droughts sit far from the playoff boundary.

The expected-pick column is where the tail channel of §4.4 becomes visible, and it is why this leg needed measuring. The best-placed beneficiary gains 0.48 picks under Waitlist COLA, against 0.49 under the lottery the NBA runs today and 0.34 under Classic COLA, and no mechanism's worst observed case exceeds 1.22 picks, which is the one pick of tail displacement Proposition 3 allows plus a small raffle contribution. Waitlist COLA does not open a channel the status quo lacks. It is exposed to the same channel at the same magnitude for the same reason, which is that a record decides pool entry under all three. The asymmetry between this leg and the previous one is the design's case in one line. Waitlist COLA gives up nothing on a channel worth about half a pick that needs a cooperative opponent at the playoff boundary, and closes a channel worth up to ten picks that needs nothing but losing.

What the run does and does not establish. The discriminating results are firm. Waitlist COLA's tanking column holds a 95% interval of $[-0.57, 0.33]$ that contains zero while Classic COLA's is $[1.65, 2.11]$ and the NBA's is $[0.85, 1.34]$; the Classic-minus-Waitlist contrast is Welch $t = 7.95$ with Cohen's $d = 1.62$, and NBA-minus-Waitlist is $t = 4.76$, $d = 0.97$. The causal zero is exact. Four limitations remain. Equity at the outcome does not separate. Across all nine mechanisms the drought tail sits between 11.6 and 11.8 years and the three-season recovery rate between 0.178 and 0.196, with Waitlist COLA at 0.196 ± 0.005 against the NBA lottery's 0.194 ± 0.005. That null follows from the modest draft channel, and it caps what can be claimed for equity at the outcome. Low power is not the explanation, since the intervals are tight.

The manipulation figures are opportunity envelopes from the engine's index distribution, so they bound the temptation available, leaving take-up unmeasured. The 15-season horizon partly censors the drought tail. And the exact pick model behind the counterfactual and the manipulation leg is a model of the draw, though it was validated against the engine first, reproducing all 21,600 deterministic tail assignments and matching realized pick-one frequencies within sampling error in every bin, and the causal figures were re-derived by an independently written Monte Carlo implementation of all three mechanisms that agrees to sampling error (1.31 against 1.32, 1.86 against 1.87, zero against zero).

The configuration that clears all three legs. Waitlist COLA holds the tanking column at zero, concentrates help on the long-suffering as hard as any configuration that also holds that column at zero, and concedes no more to pool manipulation than the lottery the league runs today. It asks one change of Classic COLA: order picks 5 through 14 by the carry-over index, and break ties on that index at random. Everything a broadcast audience recognizes survives, since the four-pick raffle on stage is untouched and its odds come from the same index Classic already uses. What does not survive is the part of the draft that pays a team for losing, which turns out to have been living entirely in those ten unlotteried slots. Two steeper weightings clear both columns as well, squaring the index reaching $+1.77$ and cubing it $+2.43$ on column (b), and we report them as viable. Both buy that concentration by distorting the raffle. Waitlist COLA reaches the same place with odds left exactly proportional to how long each team has waited, by putting the sharpness in the deterministic tail. [**Q-HIGHLEY:** Two presentational calls are yours. Should Classic COLA stay in the tables as a reference row given your read that it is clearly inferior to other family members, or drop out and leave the NBA lottery as the sole status-quo baseline? And do you want Waitlist COLA presented as a distinct named variant, as it is here, or as a setting of Classic COLA?]

5.4 Historical Backtest

The backtest computes COLA state for every team across 26 NBA seasons (1999–2025) using data verified against Basketball Reference, the standard public archive of NBA results: 775 team-season records, 776 first-round draft picks, and 55 lottery-day-attributed pre-draft pick trades. The engine separates pre-draft state (Phase A, used for draft-probability calculations) from post-draft diminishment (Phase B, carried forward to the next season). Cold-start effects stabilize by approximately season 5; we report aggregates over 21 stable seasons (2005–2025). The backtest is *static*: it applies COLA rules to actual historical outcomes. Under a real COLA regime, draft outcomes would differ in ways that change future drought lengths and indices. We caveat the comparative cross-variant claims accordingly. An audit of all 364 lottery picks (14 picks $\times$ 26 seasons) identified 8 instances where the draft-day holder differed from the lottery-day holder via post-lottery trades; COLA uses *lottery-day* attribution.

The 2013–14 through 2016–17 Philadelphia 76ers are the canonical strategic-tanking sequence in modern NBA history: four consecutive top-3 lottery outcomes (picks #3, #3, #1, #3) acquired through deliberate multi-season non-competitiveness. Under the current weighted lottery, each tanked season yielded a top-tier pick. Under Classic COLA, graduated diminishment makes the same sequence self-defeating. Philadelphia's Classic COLA trajectory: 2,750 in 2013–14 (rank 8/14) $\rightarrow$ 2,375 in 2014–15 (9/14) $\rightarrow$ 2,188 in 2015–16 (10/14) $\rightarrow$ 1,000 in 2016–17 (14/14). Each top-3 pick triggers graduated diminishment (#1: full reset; #3: 50% reduction), leaving Philadelphia at the bottom of the lottery pool by the fourth tanking season despite a 10–72 record. The Sacramento Kings hold the top Classic COLA index throughout this period, climbing from 7,250 in 2013–14 to 10,250 in 2016–17 via organic non-competitiveness. The Classic diminishment schedule ($\{1.0, 0.75, 0.5, 0.25\}$ for picks #1–4) dominates the $\alpha = 1{,}000$ annual accumulation; a team that receives a top-3 pick in year t cannot re-enter the top of the lottery in year $t + 1$ through tanking alone.

The Capped cap calibration sweep over MAX $\in \{75, 100, 125, 150, 175, 200\}$ across the 21 stable seasons replicates the published default of MAX $= 150$ [10]: at MAX $= 150$, a mean of 1.62 teams sit at the cap (range 0–3), matching the design target of fewer than two teams at the maximum on average. Sub-100 caps over-compress the rank-1-vs-rank-5 separation; caps above 175 leave fewer than one team at the cap on average. The static backtest is a second-tier validation alongside the headline full-engine result, confirming under historical data the same comparative statics the full-engine study reports under closed-loop dynamics. Trade handling is a practical implementation question for any deployment; following Highley [11], we recommend that diminishment follow the team that originally held the pick, paired with a strengthened version of the Stepien Rule, the existing NBA provision barring a team from trading away its first-round pick in consecutive future drafts.

6 Discussion and Open Directions

The result this paper reports is narrow and, we think, useful because of it. The tanking reward in a COLA draft does not live in the lottery odds, the increment schedule, the diminishment schedule, or the size of the eligible pool. It lives in the ten picks that are handed out after the televised part is over, and it disappears when those ten picks are ordered by the same multi-year index the raffle already uses. Every mechanism we tested that draws or ranks its tail on a record pays for losing, and every mechanism that does neither does not.

Two consequences follow for how the family should be presented. The first concerns the four-pick raffle, which is the part of the lottery a general audience knows and the part the league has repeatedly re-tuned. Once the raffle draws on the carry-over index, it stops rewarding a losing season, and what is left to fix sits entirely in the ten picks behind it. Waitlist COLA keeps the raffle untouched and changes only those ten. The second is that assumption (3) of the incentive-compatibility argument, that positions 5 through 14 do not matter enough at the margin to be worth manipulating, can be retired. §5.3 prices that margin at 1.87 picks under Classic COLA, and the configuration that orders those positions by the index dispenses with the assumption entirely.

Theorem 2 bounds the pool-manipulation channel at approximately 4% under empirically calibrated parameters, with explicit sensitivity to $T_{\max}$, T_{boundary}, n, and $\bar{k}$; Proposition 3 bounds the tail displacement a pool swap can cause at one pick under any deterministic ordering, including the one the NBA uses now; and Lemma 4 bounds Capped's per-series exposure. Taken together, the manipulation channels that survive under a COLA configuration compose to a small joint exposure relative to the status-quo lottery, where repeat tanking has memoryless returns.

6.1 3-2-1 as 2026 Stopgap, COLA as 2029 Candidate

We read 3-2-1 as a short-horizon stopgap. It accepts the Banchio-Munro tradeoff by flattening odds, with the bottom three teams given fewer balls than the middle of the pool. The proposal's three-year re-evaluation provision reopens the design discussion in 2029. COLA is the natural 2029 candidate on a different axis, since dial S (carry-over scope) shifts the priority basis to multi-year playoff history, an option neither 3-2-1 nor any single-season mechanism considers. 3-2-1 sacrifices differential reward for non-tanking; COLA preserves differential reward by relocating it to a non-manipulable basis. The 2029 discussion benefits from having both options analytically characterized.

6.2 Dial-Space Adoption Tradeoffs

Each dial matters more to some audiences than others. *Fans and media* care most about E and W: the current 14-team weighted lottery fits a 30-second broadcast graphic, while Capped's exclusion-rule pool and Countdown's nested Monte Carlo do not. *General managers and owners* care most about Δ and ρ: a multi-year rebuild needs a forecastable trajectory. Classic's flat $\Delta = \alpha$ and graduated ρ are the most tractable; Capped's wins-based Δ and six-step ρ add complexity but remain forecastable; Countdown's $\Delta = d \cdot w$ multiplies factors that move in opposite directions for a rebuilding team and produces noisier multi-year forecasts. *League counsel* cares most about S: multi-year scopes raise questions about mid-season team relocation, ownership change, and league realignment that single-season scopes avoid. The NBA's apparent preference for single-season scopes in 3-2-1 is consistent with S being the dial under the most external pressure.

6.3 Reset-on-Championship as a Policy Lever

The configuration of Definition 2 exposes a third option for S beyond time-bounded windows and unbounded accumulation: an outcome-triggered reset (the *reset-on-event* value of the carry-over-scope dial). For league counsel, reset-on-championship bounds the carry-over horizon without imposing a calendar deadline, and the trigger event is already part of the league's annual cadence. For GMs, it preserves the multi-year rebuild trajectory until the rebuild succeeds; winning the championship resets the priority index. Event-based scopes extend the design space beyond time-based alternatives; a full-engine evaluation of the carry-over-scope dial, which the present study holds fixed while it varies the weighting, is the natural next sweep.

Open directions. Four questions extend this work. The first is a tighter manipulation bound. The structural anti-correlation of Definition 3 is a calibrated assumption, and deriving it from the mechanism itself remains open. A stationarity argument bounding T_{boundary} from league parameters (team count, season length, playoff structure) would close the pool-manipulation assumption entirely. The second is the pick-value gap. Our bounds address the probability a manipulator can shift. What that shift is worth is a separate quantity, and pairing the bounds with a bound on $d_k - d_{k+1}$ derived from the distribution of draft-eligible prospect quality would close the incentive in absolute terms. The third is what a "worst team" should mean. Every result here defines priority by years since a conference final, which is one choice among several defensible ones (years since a playoff series win, since a title, since a winning season), and each choice sets a different bar for what counts as success. The framework is indifferent to which is used, so the family of definitions and their comparative manipulability is a self-contained question worth its own treatment. The fourth is axiomatic embedding. The lift from Coreno and Balbuzanov [4]'s three axioms, respect for priority, envy-freeness up to one object, and strategy-proofness, to multi-period priority mechanisms remains open, and a formal lift would place COLA in the assignment-mechanism literature and clarify which one-period impossibilities transfer.

Reproducibility. The simulation harness, the sweep runners, and one analysis script per reported table are public [12]. Every empirical quantity here regenerates from that repository: league generation and every draw are seeded per replicate, so a given configuration and seed reproduce a run exactly, and the mechanisms under test are configurations of the public engine's own lottery code. Two further artifacts are available to readers who want to check the results without running anything. An interactive backtester implements all five published variants plus the 3-2-1 proposal across 26 NBA seasons [13]. A companion results page runs the current NBA lottery, Classic COLA, and Waitlist COLA side by side on a scenario the reader sets, and applies the counterfactual of §5.3 on demand, so the zero in Table 5 can be reproduced by hand.

What this changes. The NBA has adjusted its draft lottery repeatedly, and each adjustment has been an argument about odds. The measurement in §5.3 says the argument has been about the smaller half of the mechanism, and that the ten picks nobody has been arguing about carry more of the incentive than the four that are drawn on television. A league that wants the reward for losing gone does not need a new lottery.

It needs to stop assigning the picks after the lottery by record, and the carry-over index gives it something defensible to assign them by instead. That change is small enough to explain in one sentence to a broadcast audience, it leaves the drawn picks and their odds exactly as they are, and on the evidence here it takes the reward for losing to zero while directing more help to long-suffering franchises than the rule it replaces.

References

[1] Martino Banchio and Evan Munro. A no-tanking draft allocation policy. In *MIT Sloan Sports Analytics Conference*, 2020. URL https://cdn.prod.website-files.com/68d6be744d7efccc2207f571/68d6be744d7efccc2207f987_munro_no_tanking_draft_policy.pdf.

[2] Timothy Highley, Tannah Duncan, and Ilia Volkov. Carry-over lottery allocation. *arXiv preprint arXiv:2602.02487*, 2026. URL https://arxiv.org/abs/2602.02487.

[3] Timothy Highley. A Professional Draft Reform Researcher Reacts to the NBA's 3-2-1 Proposal. https://highleytj.substack.com/p/a-professional-draft-reform-researcher, 2026. Substack post grading the NBA's 3-2-1 reform proposal.

[4] Julian Coreno and Ivan Balbuzanov. Axiomatic characterizations of draft rules. *arXiv preprint arXiv:2204.08300*, 2022. URL https://arxiv.org/abs/2204.08300.

[5] Lars-Gunnar Svensson. Queue allocation of indivisible goods. *Social Choice and Welfare*, 11(4), 1994.

[6] Richard J. Lipton, Evangelos Markakis, Elchanan Mossel, and Amin Saberi. On approximately fair allocations of indivisible goods. In *Proceedings of the 5th ACM Conference on Electronic Commerce (EC)*, pages 125–131, 2004.

[7] Eric Budish. The combinatorial assignment problem: Approximate competitive equilibrium from equal incomes. *Journal of Political Economy*, 119(6):1061–1103, 2011.

[8] Timothy Highley. NBA Tanking Is Solvable: Here's Why It's Hard. https://highleytj.substack.com/p/nba-tanking-is-solvable-heres-why, 2026. Substack Part 1.

[9] Timothy Highley. NBA Tanking Is Solvable: Four Candidates. https://highleytj.substack.com/p/nba-tanking-is-solvable-four-candidates, 2026. Substack Part 3.

[10] Timothy Highley. Overcoming Inertia to Stop NBA Tanking. https://highleytj.substack.com/p/overcoming-inertia-to-stop-nba-tanking, 2026. Substack post introducing Capped COLA.

[11] Timothy Highley. NBA Tanking Is Solvable: Bring Your Doubts. https://highleytj.substack.com/p/nba-tanking-is-solvable-bring-your, 2026. Substack Part 4.

[12] Kaushik Rajan and Timothy Highley. Simulation harness and analysis code for the Carry-Over Lottery Allocation study. https://github.com/kvr06-ai/cola-sloan-ssac27, 2026.

[13] Kaushik Rajan. COLA Explorer: Interactive Backtester for the Carry-Over Lottery Allocation Family. https://kvr06-ai.github.io/cola-manipulation-bound/, 2026. Source: https://github.com/kvr06-ai/cola-manipulation-bound.